Milestone 2: 50% Progress Report

Project: Real-Time and Video-Based Motion Detection and Activity Monitoring System
Team: Alabat Cindy I · Porcopio Zheniel Ivan · Veridiano Wilmark — CSST 106, Jeremy C. Reyes
Repository: https://github.com/Wilul1/Computer_Vision (tag `cp2-50`) · **Date:** October 29, 2026

1. Task status

Task (from proposal)	Owner	Status	Evidence
Python environment, pinned requirements, README setup	Wilmark	Done	`requirements.txt`, `README.md` (tested on a fresh copy)
Webcam / video / image input module	Wilmark	Done	`src/predict.py`: one command for a file or camera index
Preprocessing: resize, CLAHE, blur	Cindy	Done	`src/preprocess.py`, `docs/figures/preprocessing.jpg`
Motion baseline: MOG2 + contours, event counter	Cindy	Code done, count pending	`src/motion.py`, `scripts/eval_motion_baseline.py`
Person detection: pretrained YOLO11n	Zheniel	Done	`src/detector.py`
Pose estimation: MediaPipe, 33 landmarks	Zheniel	Done	`src/pose.py`
Features: joint angles, thigh ratio, torso tilt, speeds	Wilmark	Done	`src/features.py`
Rule-based classifier + temporal smoothing	Wilmark	First version	`src/classifier.py`, thresholds in `src/config.py`
Overlay UI: boxes, skeleton, label, FPS	Cindy	Done	`src/visualize.py`
Public dataset: import, split by source, data card	Zheniel	Done (walking, running)	`scripts/import_public.py`, `data/splits.csv`, `data/README.md`
Own recordings: sitting, standing, lying	All	In progress	`scripts/record_clips.py`
Evaluation scripts + val numbers	Wilmark	Done	`scripts/run_all_eval.py`, `outputs/metrics/activity_val.json`
Design justification	All	Done	`docs/design_choices.md`
Threshold tuning, camera-motion fix, fall detection, lighting tests	All	Not started (Phase 3–4)	

2. Dataset

294 clips from **HMDB51** (walk and run classes), restricted to full-body, single-person, non-bad-quality clips with at most 3 clips per movie. The split is **by source movie** (119 movies), so no movie appears in two splits. The test split is locked in `data/splits.csv` and was not used for anything in this report.

Split	Walking	Running	Clips	Source movies
train	127	78	205	79
val	32	12	44	20
test	27	18	45	20

All clips are 240 px tall and 30 fps, with a median length of 2.7 s (15 minutes in total). The sitting, standing and lying classes come from our own webcam and phone recordings, which are in progress. HMDB51's "sit" and "stand" classes are the *motions* of sitting down and standing up, not postures, so we did not use them.

3. Preprocessing



Columns: original → resized (long side ≤ 640) → CLAHE on the Lab L-channel → grayscale + 5×5 Gaussian (MOG2 input) → MOG2 mask → pose landmarks with the extracted features. Rows 3–4 show that **MOG2 produces scattered noise when the camera moves**, as it does in most movie clips. Background subtraction assumes a fixed camera. The reasons for each setting are in `docs/design_choices.md`.

4. Baseline numbers (validation split)

Reproduce with `python scripts/eval_activity.py --split val`, or `python scripts/run_all_eval.py`.

Method	Metric	Value
Pose + rules (main method)	Clip-level accuracy, 44 val clips	**0.477**
Pose + rules (main method)	Macro-F1 (walking, running)	**0.510**
Full pipeline (YOLO + pose + MOG2 + rules)	Processing speed, laptop CPU (i7-6700HQ), 240p clips	**11.8 FPS** (7–9 FPS on 640×480 webcam)
MOG2 + contours (motion baseline)	Motion events vs hand count	Pending (see below)

Confusion matrix (rows = true, columns = predicted):

true \ predicted	walking	running	standing	sitting	no person
walking (32)	**18**	2	11	0	1
running (12)	4	**3**	3	2	0

What the errors tell us (first look; full error analysis comes in Phase 3):

1. **Walking → standing (11 clips).** In movies the camera usually *follows* the actor, so the person stays in the same place in the frame. Our hip speed is measured relative to the frame, so it is close to zero. Planned fix: measure speed relative to the background (optical flow on background points), and rely more on leg swing.
2. **Running → walking or standing (7 clips).** This has the same camera-following cause, plus short clips (median 2.7 s) where the 9-frame smoothing and 0.5 s speed window use up a large part of the clip.
3. **Running → sitting (2 clips).** During a running stride the knee bends below 130°. When the speed is underestimated (cause 1), this matches the sitting rule. Planned fix: require the bent posture to last at least 1 s before calling it sitting.

Motion baseline count: the hand-count comparison needs a **fixed camera**. On HMDB's moving-camera clips the MOG2 mask is mostly background noise (figure rows 3–4), so a count there would not be meaningful. It will be measured on 3 static-camera recordings of our own, as soon as they are recorded.

5. Percent complete

Work package (whole project)	Weight	Done
Environment, input module	5	5
Preprocessing	5	5
MOG2 motion baseline (code + count)	5	4
YOLO + pose integration	10	10
Features + rule classifier (first version)	15	12
Overlay UI	5	5
Dataset (public + own recordings)	10	5
Evaluation scripts + val numbers	10	9
Threshold tuning + camera-motion fix	15	0
Fall detection	5	0
Lighting tests + final test-split evaluation	5	0
Documentation, presentation, demo	10	3
Total	**100**	**58**

58% complete. The weights are our estimate of effort. "Done" counts only work that runs and is in the repository.

6. Blockers

- **Own recordings** for sitting, standing and lying, and static-camera clips for the motion count. Each member records 2 sessions in different rooms and lighting.
- **Moving cameras in public data** make frame-relative speed unreliable (error causes 1–2 above).

7. Next dates

- **Oct 31 – Nov 1:** own recordings done, added to the split as new sources (the existing test split does not change), motion hand count measured.
- **Nov 2 – Nov 12 (80% check):** tune thresholds on val, add camera-motion compensation, minimum-duration rule for sitting, fall detection, tests in different lighting.
- **Nov 13–20:** final evaluation on the locked test split, presentation and live demo.

8. Correction to the proposal timeline

Weeks 3–5 of the proposal mention squats, push-ups, lunges, jumping jacks and repetition counting. Those were carried over by mistake. The project classifies standing, sitting, walking, running and lying, as stated in the objectives.